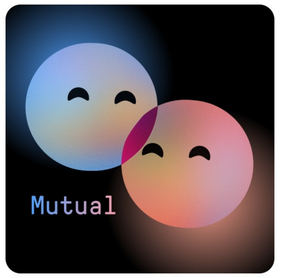
CAL HACKS 13.0 · HCI · UC BERKELEY, OCTOBER 2026

Human Human Interaction Lab

Filtering the crowd so every hello counts: a reciprocal, proximity aware name badge on the phone you already own

Ethan Pan · Paul Jiang · Jiayi Tang · Kenny Nguyen

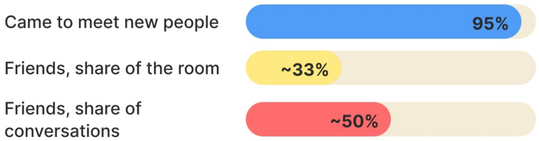
UC Berkeley · UC Irvine · Simon Fraser University



1 The problem

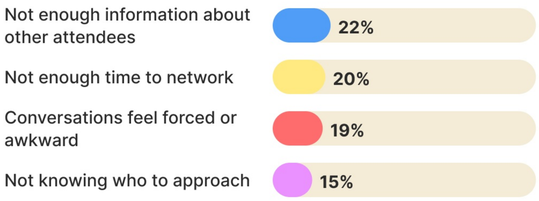
Hackathons, conferences, career fairs, orientation: you get a few hours to find a team, a cofounder or a mentor among hundreds of strangers. Name tags say nothing about fit, so every conversation is a gamble with your limited time, and the right person may be ten feet away.

People don't mix at mixers



Ingram and Morris, Administrative Science Quarterly, 2007. ~100 executives wearing electronic name tags at a mixer.

Why networking at events falls short



Each attendee named their single biggest challenge; these four add up to 76%. Bizzabo 2025 Event Networking Report, survey of 300+ event organizers, attendees and sponsors. Industry survey.

38% of professionals find it hard to stay in touch with their network. The top reason: not enough time.

People value their contacts, but few keep them up:



LinkedIn global survey of 15,905 members in 17 countries, Feb to Mar 2017 (LinkedIn Pressroom, June 2017). The top reason was cited by 49%. Industry survey.

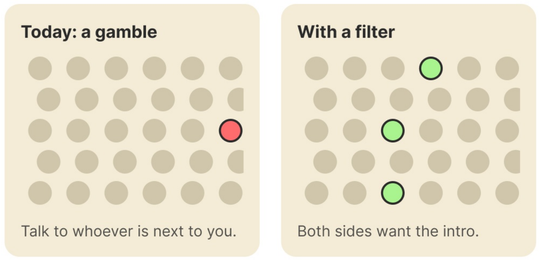
2 Our thinking

THE QUESTION

Can we make each encounter more valuable on average?

Not more conversations. Better odds for each one.

Filtering beats picking at random



Evidence: meeting the right people is the bottleneck. Researchers randomly placed in one shared 90 minute session were 75% more likely to apply for a grant together (Boudreau et al., Review of Economics and Statistics 2017). When a hurricane cancelled APSA 2012, would be attendees became 16% less likely to coauthor (Campos et al., Economic Journal 2018).

What we believe

- **Filter before you talk**
  Reciprocal: an intro only happens when both people would want it.
- **Keep eyes up**
  The signal is worn on your chest, not buried in another feed.
- **Lower the fear**
  A shared color says: the other person wants this too.
- **Close the loop**
  A summary and a follow up arrive by text after you talk.
- **No new hardware**
  The phone you own, in a 3D printed lanyard holder.


Cal Hacks 13.0: hundreds of hackers, one hangar

3 Related work

MIT Media Lab badges

Borovoy et al. 1998 · Paradiso et al. 2010

Did: Meme Tags swapped short ideas face to face; later badges sensed who met whom.

Inspired: give strangers something to talk about.

Different: no custom hardware; topics flip to what this person wants to discuss with you.


Our drawing · Meme Tag, 1998 · UbER-Badge, 2010 · Communicator badge

Photos from the papers: Borovoy et al., CSCW 1998, Fig. 1; Paradiso et al., Personal and Ubiquitous Computing 2010, Figs. 1 and 18 (MIT Media Lab).

Social Serendipity

Eagle and Pentland, published in IEEE Pervasive Computing 2005

Did: Bluetooth phones alerted two nearby people whose profiles matched.

Inspired: an intro needs both sides to clear the bar.

Different: an LLM scores needs in plain language, and the cue is worn, not a phone alert.


Figure from Eagle and Pentland, IEEE Pervasive Computing 2005

SocialStools

Guo, Leshed and Green, published at CHI 2023

Did: stools whose distance drives shared sound and light between strangers.

Inspired: map physical distance to a shared cue.

Different: our cue is a color both badges take on, only for mutual matches.


Our drawing of distance as a shared cue

RECON

Pizzato et al., published at RecSys 2010

Did: a reciprocal recommender for online dating.

Inspired: score each direction of interest.

Different: a geometric mean of LLM fit, used in the physical room.

What's new here

- Matching that needs both sides, scored by an LLM from natural language
- A physical, ambient signal you wear, not a phone alert
- Peer to peer UWB: no venue readers, no custom badge
- The loop closes after the chat: summary, follow up, a personal CRM

Next: measure intros per hour, and how many saved contacts become a real follow up.

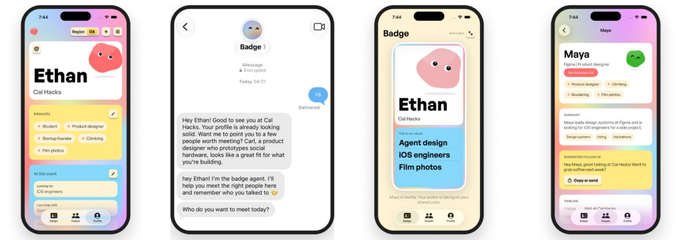
4 Our solution

A live badge you wear on a lanyard



Worn on a lanyard (AI rendering) · Alone: your name and topics · Match nearby: shared tint · Talking: avatars hop in

How it works



1 Onboard — Text Photon what you want and offer. Claude builds your profile.

2 Heads up — Pairs are scored both ways. Only mutual fits get a text.

3 Walk closer — UWB distance tints both avatars toward one shared color.

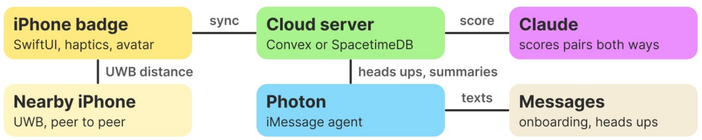
4 Follow up — Then a summary and follow up text. People keeps who you met.

RECIPROCAL SCORE, COMPUTED BY CLAUDE FOR EVERY PAIR

$$\text{score} = 0.75\sqrt{\text{fit}_{\text{me}} \cdot \text{fit}_{\text{them}}} + 0.25 \text{ values}$$

A geometric mean punishes one sided interest: fits of 0.9 and 0.1 score 0.30, below 0.5 and 0.5 at 0.50.

Architecture



References: Ingram & Morris, Administrative Science Quarterly 2007 · Bizzabo, 2025 Event Networking Report · LinkedIn global networking survey, 2017 · Boudreau et al., Review of Economics and Statistics 2017 · Campos, Leon & McQuillin, Economic Journal 2018 · Borovoy et al., CSCW 1998 · Paradiso et al., Personal and Ubiquitous Computing 2010 · Eagle & Pentland, IEEE Pervasive Computing 2005 · Guo, Leshed & Green, CHI 2023 · Pizzato et al., RecSys 2010. Worn badge photo is an AI rendering made with ChatGPT.

Built at Cal Hacks 13.0 with Convex, Claude, Photon and Figma.

Demo and Devpost: devpost.com/software/human-human-interaction-lab